

Project: StayNest

Hotel & Hospitality Management System

Domain: Travel & Hospitality

1. Introduction

StayNest is a web-based hotel and hospitality management system designed for hotel chains, boutique properties, and hospitality groups to manage room inventory, guest reservations, front-desk operations, housekeeping workflows, food & beverage billing, and guest experience analytics. The system enables guests to browse availability and manage bookings, front-desk staff to handle check-in/check-out and billing, housekeeping supervisors to manage room readiness, and administrators to configure room types, tariffs, and service menus.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

Actors / Users

- **Guest:** Searches availability, makes reservations, manages stays, and provides feedback
- **Front Desk Agent:** Handles check-in, check-out, room assignments, and billing
- **Housekeeping Supervisor:** Assigns cleaning tasks, tracks room readiness, and manages linen inventory
- **F&B Manager:** Manages menu items, dining reservations, and in-room dining orders
- **Revenue Manager:** Monitors occupancy, manages dynamic tariff rules, and reviews revenue reports
- **Hotel Admin:** Configures room types, tariffs, amenities, service menus, and user roles

2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.
- **2.2 Room Inventory & Configuration:** Manages room types, floor plans, amenities, and availability calendars.
- **2.3 Reservation & Booking Management:** Handles guest reservations, modifications, and cancellations.

- **2.4 Front Desk & Stay Management:** Manages check-in, check-out, room assignments, and guest folios.
- **2.5 Housekeeping Management:** Tracks room cleaning tasks, statuses, and maintenance requests.
- **2.6 F&B & Ancillary Services:** Manages dining menus, orders, and in-room service billing.
- **2.7 Revenue & Hospitality Analytics:** Provides dashboards for occupancy rates, ADR, RevPAR, and guest satisfaction.
- **2.8 Notifications & Alerts:** Booking confirmations, check-in reminders, housekeeping alerts, and guest feedback prompts.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features

- Staff and guest account registration, login, and role-based access control
- Shift-based session management and full audit logging

Entities

- **User**
 - UserID
 - Name
 - Role (Guest/FrontDesk/Housekeeping/FBManager/RevenueManager/Admin)
 - Email
 - Phone
 - PropertyID
 - Status (Active/Inactive)

- **AuditLog**
 - AuditID
 - UserID
 - Action
 - EntityType
 - Timestamp

4.2 Room Inventory & Configuration Module

Features

- Define room types, floor assignments, bed configurations, and amenities
- Manage availability calendars and room blocking for maintenance

Entities

- **RoomType**
 - RoomTypeID
 - Name (Standard/Deluxe/Suite/Villa)
 - BedConfiguration
 - MaxOccupancy
 - BaseRate
 - AmenitiesList
 - Status (Active/Inactive)
- **Room**
 - RoomID
 - RoomNumber
 - Floor
 - RoomTypeID
 - PropertyID
 - Status (Available/Occupied/Cleaning/Maintenance/Blocked)
- **RatePlan**

- RatePlanID
- RoomTypeID
- Name (Rack/Corporate/Seasonal/Promo)
- PricePerNight
- ValidFrom
- ValidTo
- MealPlanIncluded
- Status

4.3 Reservation & Booking Management Module

Features

- Create and manage guest reservations across direct and walk-in channels
- Handle booking modifications, cancellations, and no-show tracking

Entities

- **Reservation**
 - ReservationID
 - GuestID
 - RoomTypeID
 - RatePlanID
 - CheckInDate
 - CheckOutDate
 - Nights
 - Adults
 - Children
 - TotalAmount
 - BookingChannel (Direct/Phone/Walkin)
 - Status (Confirmed/CheckedIn/CheckedOut/Cancelled/NoShow)
- **GuestProfile**

- GuestID
- Name
- Email
- Phone
- Nationality
- IDDocumentType
- IDNumber
- PreferencesJSON
- LoyaltyTier (None/Silver/Gold/Platinum)
- Status (Active/Blacklisted)

4.4 Front Desk & Stay Management Module

Features

- Process arrivals and assign rooms based on availability and preferences
- Maintain real-time guest folios with itemised charges and checkout billing

Entities

- **StayRecord**
 - StayID
 - ReservationID
 - GuestID
 - AssignedRoomID
 - ActualCheckIn
 - ActualCheckOut
 - FolioBalance
 - Status (Active/CheckedOut)
- **FolioItem**
 - FolioItemID
 - StayID

- ChargeType (RoomRent/FBCharge/LaundryCharge/Spa/Tax/Discount)
- Description
- Amount
- PostedDate
- PostedBy

4.5 Housekeeping Management Module

Features

- Generate daily housekeeping task lists based on checkouts and stay-overs
- Track room cleaning status and manage maintenance work orders

Entities

- **HousekeepingTask**
 - TaskID
 - RoomID
 - TaskType (Checkout/StayoverService/TurnDown/DeepClean)
 - AssignedToID
 - AssignedDate
 - CompletedAt
 - Status (Pending/InProgress/Done/Skipped)
- **MaintenanceRequest**
 - RequestID
 - RoomID
 - ReportedBy
 - IssueDescription
 - Priority (Low/Medium/High/Urgent)
 - RaisedDate
 - ResolvedDate
 - Status (Open/InProgress/Resolved/Deferred)

4.6 F&B & Ancillary Services Module

Features

- Manage restaurant menus, table reservations, and point-of-sale orders
- Handle in-room dining orders and post charges to guest folios

Entities

- **MenuItem**
 - MenuItemID
 - Name
 - Category (Breakfast/MainCourse/Beverage/Dessert)
 - Price
 - IsAvailable
 - DietaryTags
- **FBOOrder**
 - OrderID
 - StayID
 - TableNumber
 - OrderType (DineIn/InRoomDining/Takeaway)
 - ItemsJSON
 - TotalAmount
 - OrderTime
 - Status (Placed/Preparing/Served/Billed/Cancelled)
- **DiningReservation**
 - DiningResID
 - GuestID
 - StayID
 - RestaurantOutlet
 - Date

- Time
- Covers
- Status (Confirmed/Seated/Completed/Cancelled/NoShow)

4.7 Revenue & Hospitality Analytics Module

Features

- Occupancy rate, ADR, and RevPAR dashboards by property, room type, and period
- Housekeeping efficiency, F&B revenue, and guest satisfaction score metrics

Entities

- **HospitalityReport**
 - ReportID
 - Scope (Property/RoomType/Period/Channel)
 - Metrics (OccupancyRate, ADR, RevPAR, AvgLengthOfStay, FBRevenue, GuestSatisfactionScore)
 - GeneratedDate

4.8 Notifications & Alerts Module

Features

- Booking confirmation, pre-arrival reminders, and checkout prompts for guests
- Housekeeping task alerts, maintenance escalations, and occupancy threshold warnings for staff

Entities

- **Notification**
 - NotificationID
 - UserID
 - Message
 - Category (Reservation/FrontDesk/Housekeeping/FB/Revenue)
 - Status (Unread/Read/Dismissed)
 - CreatedDate

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB
- **Production:** Cloud or on-premises with secure APIs and stateless services

6. Database Design

Tables:

- **User**(UserID, Name, Role, Email, Phone, PropertyID, Status)
- **AuditLog**(AuditID, UserID, Action, EntityType, Timestamp)
- **RoomType**(RoomTypeID, Name, BedConfiguration, MaxOccupancy, BaseRate, AmenitiesList, Status)
- **Room**(RoomID, RoomNumber, Floor, RoomTypeID, PropertyID, Status)
- **RatePlan**(RatePlanID, RoomTypeID, Name, PricePerNight, ValidFrom, ValidTo, MealPlanIncluded, Status)
- **GuestProfile**(GuestID, Name, Email, Phone, Nationality, IDDocumentType, IDNumber, PreferencesJSON, LoyaltyTier, Status)
- **Reservation**(ReservationID, GuestID, RoomTypeID, RatePlanID, CheckInDate, CheckOutDate, Nights, Adults, Children, TotalAmount, BookingChannel, Status)
- **StayRecord**(StayID, ReservationID, GuestID, AssignedRoomID, ActualCheckIn, ActualCheckOut, FolioBalance, Status)
- **FolioItem**(FolioItemID, StayID, ChargeType, Description, Amount, PostedDate, PostedBy)
- **HousekeepingTask**(TaskID, RoomID, TaskType, AssignedToID, AssignedDate, CompletedAt, Status)
- **MaintenanceRequest**(RequestID, RoomID, ReportedBy, IssueDescription, Priority, RaisedDate, ResolvedDate, Status)
- **MenuItem**(MenuItemID, Name, Category, Price, IsAvailable, DietaryTags)
- **FBOOrder**(OrderID, StayID, TableNumber, OrderType, ItemsJSON, TotalAmount, OrderTime, Status)
- **DiningReservation**(DiningResID, GuestID, StayID, RestaurantOutlet, Date, Time, Covers, Status)
- **HospitalityReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Guest Booking Portal:** Room availability search, reservation management, stay history, feedback submission
- **Front Desk Console:** Arrivals and departures dashboard, room assignment grid, guest folio manager
- **Housekeeping Dashboard:** Daily task list, room status board, maintenance request tracker
- **F&B POS & Reservations:** Menu builder, order entry, dining reservation calendar, folio posting
- **Revenue Manager Dashboard:** Occupancy heat map, ADR and RevPAR trends, rate plan performance
- **Admin Console:** Property configuration, room type setup, rate plan management, user roles

8. Non-Functional Requirements

- **Performance:** Support 10,000+ concurrent users across properties including front desk and mobile devices
- **Security:** Guest PII protection, secure folio data, RBAC, and full audit trails for financial postings
- **Scalability:** Multi-property architecture with shared catalog and independent operational data per property
- **Availability:** 99.95% uptime for front desk and housekeeping modules critical to round-the-clock operations
- **Maintainability:** Configurable rate plans, housekeeping rules, and F&B menu management without code changes
- **Observability:** Real-time room status board, occupancy dashboards, and housekeeping SLA monitoring

9. Assumptions & Constraints

- Phase 1 excludes integration with external OTA channels (Booking.com, Expedia) and POS hardware
- Online payments are simulated internally; payment gateway integration is deferred to Phase 2

- All notifications are in-app only; guest-facing email/SMS integrations are out of Phase 1 scope
- Loyalty point accrual and redemption is tracked internally; partner network integration is deferred
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling